

Qatar-1b

R-0313

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_170526 · created 2026-07-17T23:34:15+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457899.7936 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.189 +/- 0.05
Transit depth (Rp/Rs)^2	3.56 +/- 1.89 [%]
Orbital Inclination (inc)	81.51 +/- 2.52
Airmass coefficient 1 (a1)	1.015 +/- 0.024
Airmass coefficient 2 (a2)	-0.0075 +/- 0.0085
Scatter in the residuals of the lightcurve fit is	2.32 %
Best Comparison Star	#3 - [196, 143]
Optimal Aperture	3.45
Optimal Annulus	10.27
Transit Duration (day)	0.047 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.189 $\pm$ 0.05 vs 0.1463 (deviation 0.59 σ)
Duration fit vs published	0.047 d vs 0.0692 d (deviation 0.57 σ)
Mid-time O-C	2.6 min
Depth SNR	2.6
Residual scatter	2.32 %

Light curve

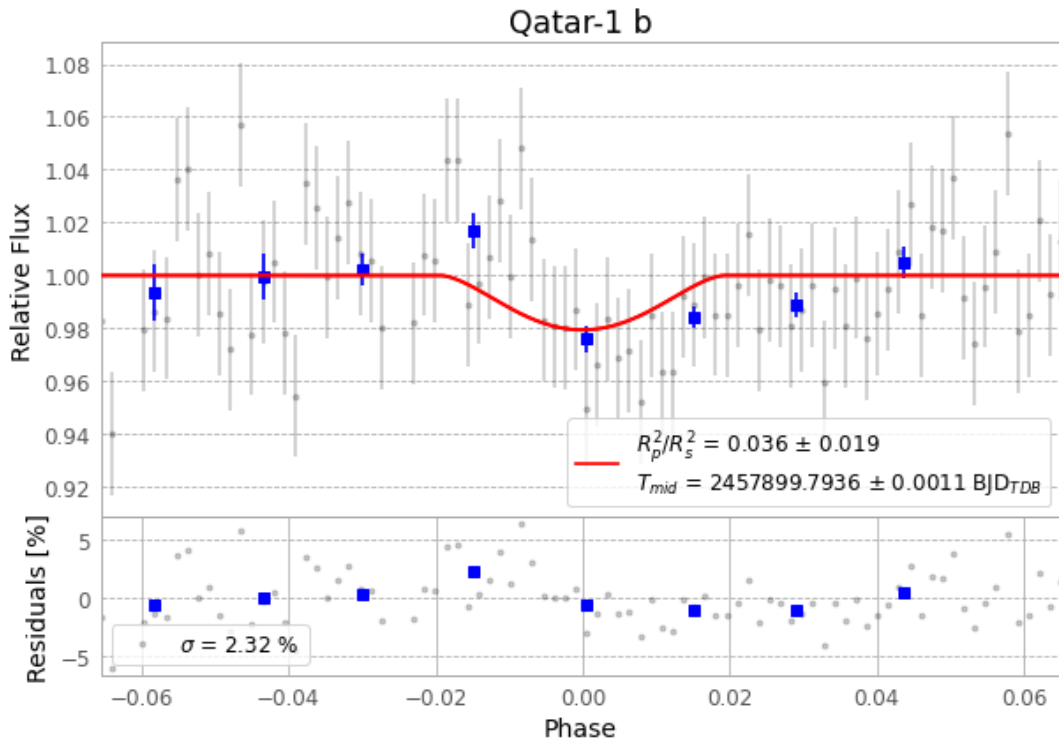


Figure 1 — FinalLightCurve_Qatar-1 b_2017-05-25.png (SHA-256 2a8741732af749ea...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [148, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b209bac3a0d43a87...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1bbef5c

Data provenance

91 FITS frames (60 MB), Qatar-1170526044512.FITS → Qatar-1170526091512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-170526.log (389b7c187ad7...), FinalLightCurve_Qatar-1 b_2017-05-25.png (2a8741732af7...), FinalLightCurve_Qatar-1 b_2017-05-25.csv (3dab51050121...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 23:34Z.